

Beginning Algebra Practice Final Test (Part II) Version 1

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show work, even when you can do the problems in your head.
- iii) Draw a rectangle around your answers.
- iv) Staple a copy of these test questions with your work and your final answers.
- v) You may use a calculator (but not a phone or a graphing calculator).

Solve.

$$\textcircled{1} \frac{(3-2x)^2}{-7} + 2 = -2$$

$$\frac{(3-2x)^2}{-7} = -4$$

$$(3-2x)^2 = 28$$

$$\sqrt{(3-2x)^2} = \pm \sqrt{28}$$

$$3-2x = \pm 2\sqrt{7}$$

$$-2x = \pm 2\sqrt{7} - 3$$

$$x = \frac{\pm 2\sqrt{7} - 3}{-2}$$

$$\boxed{x = \frac{\pm 2\sqrt{7} + 3}{2}}$$

$$\textcircled{2} \begin{cases} 2x - y = 3 \\ 5x + 3y = 24 \end{cases}$$

$$3 \cdot \begin{cases} 2x - y = 3 \\ 5x + 3y = 24 \end{cases}$$

$$\begin{cases} 6x - 3y = 9 \\ 5x + 3y = 24 \end{cases}$$

$$11x + 0 = 33$$

$$11x = 33$$

$$\boxed{x = 3}$$

$$2(3) - y = 3$$

$$\boxed{y = 3}$$

- ③ Use the slope-intercept form to find the slope-intercept form of the equation of the line with slope 4 and containing the point (2, -2).

$$y = mx + b$$

$$-2 = 4(2) + b$$

$$-2 = 8 + b$$

$$-10 = b$$

$$b = -10, m = 4$$

$$y = mx + b$$

$$\boxed{y = 4x - 10}$$

- ④ Find an equation of the line with the following characteristics.

Y-intercept: $\frac{1}{2}$

Point: (2, 5)

Method I

$$b = \frac{1}{2}$$

$$y = mx + b$$

$$5 = m(2) + \frac{1}{2}$$

$$5 = 2m + \frac{1}{2}$$

$$\frac{9}{2} = 2m$$

$$m = \frac{9}{4}$$

$$y = mx + b$$

$$\boxed{y = \frac{9}{4}x + \frac{1}{2}}$$

Method II

(2, 5) & (0, $\frac{1}{2}$)
(x_1, y_1) & (x_2, y_2)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{\frac{1}{2} - 5}{0 - 2} = \frac{9}{4}$$

$$y = mx + b$$

$$\boxed{y = \frac{9}{4}x + \frac{1}{2}}$$

⑤ Find an equation of the line with X-intercept 7 and that is perpendicular to the line with equation $-2x - 4 = y$.

Other Line

$$-2x - 4 = y$$

$$mx + b = y$$

$$m = -2$$

$$m_{\perp} = \frac{1}{2}$$

The Line We're Looking For

$$(7, 0), m_{\perp} = \frac{1}{2}$$

$$y - y_p = m(x - x_p)$$

$$y - 0 = \frac{1}{2}(x - 7)$$

⑥ Find an equation of the line that contains the point $(9, -3)$ and is parallel to the line with equation $y - 6 = -\frac{4}{5}(x + 1)$.

Other Line

$$y - 6 = -\frac{4}{5}(x + 1)$$

$$y - y_p = m(x - x_p)$$

$$m = -\frac{4}{5}$$

$$m_{\parallel} = -\frac{4}{5}$$

The Line We're Looking For

$$(9, -3), m_{\parallel} = -\frac{4}{5}$$

$$y - y_p = m(x - x_p)$$

$$y + 3 = -\frac{4}{5}(x - 9)$$

⑦ Find an equation of the line that contains the point $(-4, -6)$ and is parallel to the line with equation $-3x + 6y = 7$.

Other Line

$$-3x + 6y = 7$$

$$Ax + By = C$$

$$6y = 3x + 7$$

$$y = \frac{x}{2} + \frac{7}{6}$$

$$y = \frac{1}{2}x + \frac{7}{6}$$

$$y = mx + b$$

$$m = \frac{1}{2}$$

$$m_{\parallel} = \frac{1}{2}$$

The Line We're Looking For

$$(-4, -6), m_{\parallel} = \frac{1}{2}$$

$$y - y_p = m(x - x_p)$$

$$y + 6 = \frac{1}{2}(x + 4)$$

⑧ Convert the following equation to slope-intercept form.

$$y + 7 = 2(x - 3)$$

$$y + 7 = 2x - 6$$

$$y = 2x - 13$$

Solve.

$$\textcircled{9} \sqrt{-9+y^2} = \sqrt{8y}$$

$$(\sqrt{-9+y^2})^2 = (\sqrt{8y})^2$$

$$-9+y^2 = 8y$$

$$y^2 - 8y - 9 = 0$$

$$(y-9)(y+1) = 0$$

$$y-9=0 \text{ or } y+1=0$$

$$\boxed{y=9} \text{ or } y \neq -1$$

$$\textcircled{10} \sqrt{3x+4} - 2 = \sqrt{1+3x}$$

$$(\sqrt{3x+4} - 2)^2 = (\sqrt{1+3x})^2$$

$$(\sqrt{3x+4} - 2)(\sqrt{3x+4} - 2) = 1+3x$$

$$(\sqrt{3x+4})^2 - 2\sqrt{3x+4} - 2\sqrt{3x+4} + 4 = 1+3x$$

$$3x+4 - 4\sqrt{3x+4} + 4 = 1+3x$$

$$4 - 4\sqrt{3x+4} + 4 = 1$$

$$8 - 4\sqrt{3x+4} = 1$$

$$-4\sqrt{3x+4} = -7$$

$$\sqrt{3x+4} = \frac{7}{4}$$

$$(\sqrt{3x+4})^2 = \left(\frac{7}{4}\right)^2$$

$$3x+4 = \frac{49}{16}$$

$$3x = -\frac{15}{16}$$

$$x \neq -\frac{5}{16}$$

No Solution

$$\textcircled{11} \text{ a) } 7 = 2 + 4\sqrt{x/3 - 1}$$

$$5 = 4\sqrt{x/3 - 1}$$

$$\frac{5}{4} = \sqrt{x/3 - 1}$$

$$\left(\frac{5}{4}\right)^2 = (\sqrt{x/3 - 1})^2$$

$$\frac{25}{16} = \frac{x}{3} - 1$$

$$\frac{41}{16} = \frac{x}{3}$$

$$\boxed{\frac{123}{16} = x}$$

$$\text{b) } 2\sqrt{3x-2} = \frac{\sqrt{5x}}{3}$$

$$(2\sqrt{3x-2})^2 = \left(\frac{\sqrt{5x}}{3}\right)^2$$

$$(2\sqrt{3x-2})(2\sqrt{3x-2}) = \frac{\sqrt{5x}}{3} \cdot \frac{\sqrt{5x}}{3}$$

$$4(\sqrt{3x-2})^2 = \frac{(\sqrt{5x})^2}{9}$$

$$4(3x-2) = \frac{5x}{9}$$

$$12x - 8 = \frac{5x}{9}$$

$$\frac{12x}{1} - \frac{8}{1} = \frac{5x}{9}$$

$$\frac{108x}{9} - \frac{72}{9} = \frac{5x}{9}$$

$$\frac{108x - 72}{9} = \frac{5x}{9}$$

$$108x - 72 = 5x$$

$$-72 = -103x$$

$$\boxed{\frac{72}{103} = x}$$

⑫ Find an equation of the line through the points $(-5, -6)$ and $(8, 3)$ in

a) Point-Slope form.

$(-5, -6)$ & $(8, 3)$

(x_1, y_1) & (x_2, y_2)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{3 - (-6)}{8 - (-5)}$$

$$m = \frac{3+6}{8+5}$$

$$m = \frac{9}{13}$$

$$y - y_p = m(x - x_p)$$

$$y - 3 = \frac{9}{13}(x - 8)$$

b) Slope-intercept form.

$$y - 3 = \frac{9}{13}(x - 8)$$

$$y - 3 = \frac{9}{13}x - \frac{72}{13}$$

$$y = \frac{9x}{13} + 3 - \frac{72}{13}$$

$$y = \frac{9x}{13} - \frac{33}{13}$$

c) What is the slope of the line?

$$m = \frac{9}{13}$$

d) What is the y-intercept?

$$y = \frac{9x}{13} - \frac{33}{13}$$

$$y = mx + b$$

$$b = -\frac{33}{13}$$

⑬ a) Solve the following system using the graphical method.

$$\begin{cases} x = 1 + y \\ -2x + y = 1 \end{cases}$$

$$x = 1 + y$$

Point A

$$y = -1$$

$$x = 1 + (-1)$$

$$x = 0$$

$$(0, -1)$$

Point B

$$y = 3$$

$$x = 1 + 3$$

$$x = 4$$

$$(4, 3)$$

$$-2x + y = 1$$

Point C

$$x = 0$$

$$-2(0) + y = 1$$

$$y = 1$$

$$(0, 1)$$

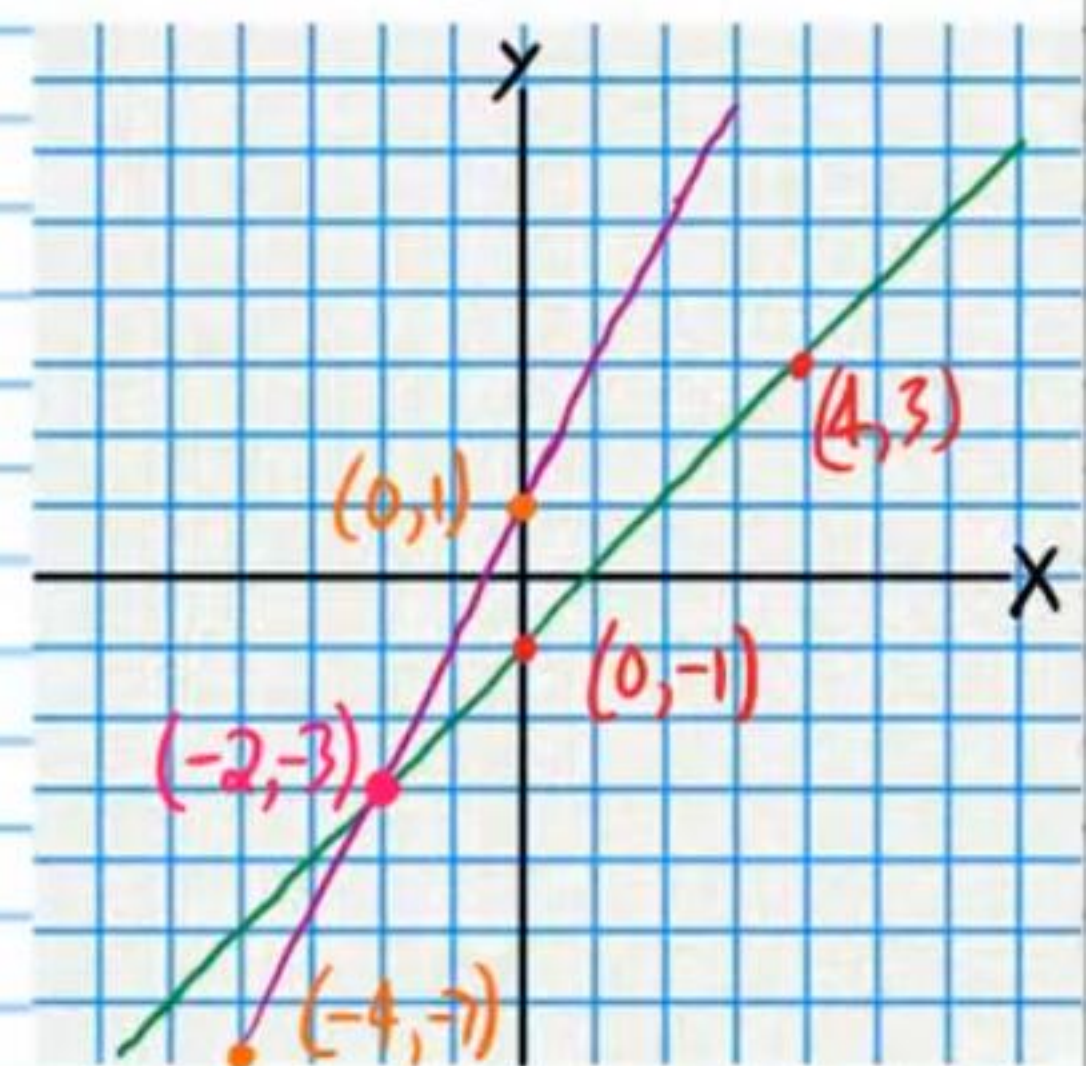
Point D

$$x = -4$$

$$-2(-4) + y = 1$$

$$y = -7$$

$$(-4, -7)$$



b) Then solve the system algebraically (i.e. Substitution or elimination) to check your solution.

$$\begin{cases} X = 1 + y \\ -2x + y = 1 \end{cases}$$

$$\rightarrow -2(1+y) + y = 1$$

$$-2 - 2y + y = 1$$

$$-2 - y = 1$$

$$\boxed{y = -3}$$

$$x = 1 + (-3)$$

$$\boxed{x = -2}$$

Solve.

$$(14) \quad 4 = 3x^2 - 23$$

$$27 = 3x^2$$

$$9 = x^2$$

$$\pm \sqrt{9} = \sqrt{x^2}$$

$$\pm 3 = x$$

$$\boxed{x = 3} \text{ or } \boxed{x = -3}$$

$$(15) \quad \frac{1}{4x-8x^2} - \frac{1}{4x} = \frac{-2x}{6x-3}$$

$$\frac{1}{4x(1-2x)} - \frac{1}{4x} = \frac{-2x}{3(2x-1)}$$

$$\frac{1}{4x(1-2x)} - \frac{1}{4x} = \frac{-2x}{-3(-2x+1)}$$

$$\frac{1}{4x(1-2x)} - \frac{1}{4x} = \frac{2x}{3(1-2x)}$$

$$\frac{3}{3} \frac{1}{4x(1-2x)} - \frac{1(3)(1-2x)}{4x(3)(1-2x)} = \frac{2x(4x)}{3(1-2x)(4x)}$$

$$\frac{3}{12x(1-2x)} - \frac{3(1-2x)}{12x(1-2x)} = \frac{8x^2}{12x(1-2x)}$$

$$\frac{3}{12x(1-2x)} - \frac{3-6x}{12x(1-2x)} = \frac{8x^2}{12x(1-2x)}$$

$$\frac{3-3+6x}{12x(1-2x)} = \frac{8x^2}{12x(1-2x)}$$

$$3-3+6x = 8x^2$$

$$6x = 8x^2$$

$$0 = 8x^2 - 6x$$

$$0 = 2x(4x-3)$$

$$2x=0 \text{ or } 4x-3=0$$

$$x \neq 0 \text{ or } \boxed{x = \frac{3}{4}}$$

$$\textcircled{16} \quad \frac{2}{5b+6} + \frac{1}{5b-6} = \frac{b+7}{25b^2-36}$$

$$\frac{2}{5b+6} + \frac{1}{5b-6} = \frac{b+7}{(5b+6)(5b-6)}$$

$$\frac{2(5b-6)}{(5b+6)(5b-6)} + \frac{1(5b+6)}{(5b-6)(5b+6)} = \frac{b+7}{(5b+6)(5b-6)}$$

$$\frac{10b-12}{(5b+6)(5b-6)} + \frac{5b+6}{(5b-6)(5b+6)} = \frac{b+7}{(5b+6)(5b-6)}$$

$$\frac{15b-6}{(5b+6)(5b-6)} = \frac{b+7}{(5b+6)(5b-6)}$$

$$15b-6 = b+7$$

$$14b = 13$$

$$\boxed{b = \frac{13}{14}}$$

$\textcircled{17}$ Gold is too soft to be used as jewelry, so it is mixed with silver. If gold costs \$1800 per ounce and silver costs \$30 per ounce, how many ounces of each metal would you have to mix to make 60 ounces of a gold and silver alloy that costs \$1003.50 per ounce?

G = # ounces of gold.

S = # ounces of silver.

$$\begin{cases} G + S = 60 \\ 1800G + 30S = 60(1003.5) \end{cases}$$

$$1800G + 30S = 60,210$$

$$\begin{cases} G + S = 60 \\ 1800G + 30S = 60,210 \end{cases}$$

$$1800G + 30S = 60,210$$

$$-30 \cdot \begin{cases} G + S = 60 \\ 1800G + 30S = 60,210 \end{cases}$$

$$1800G + 30S = 60,210$$

$$\begin{cases} -30G - 30S = -1800 \\ 1800G + 30S = 60,210 \end{cases}$$

$$1800G + 30S = 60,210$$

$$1770G + 0 = 58,410$$

$$1770G = 58,410$$

$$\boxed{G = 33 \text{ oz gold}}$$

$$33 + S = 60$$

$$\boxed{S = 27 \text{ oz silver}}$$

⑮ The Smith family and the Thompson family are going to see a show together. The Smiths bought 3 adult tickets and 5 child tickets for \$155. The Thompsons bought 4 adult tickets and 2 child tickets for \$132. What is the price per child ticket?

C = Price per child ticket

A = Price per adult ticket

$$\begin{cases} 3A + 5C = 155 \\ 4A + 2C = 132 \end{cases}$$

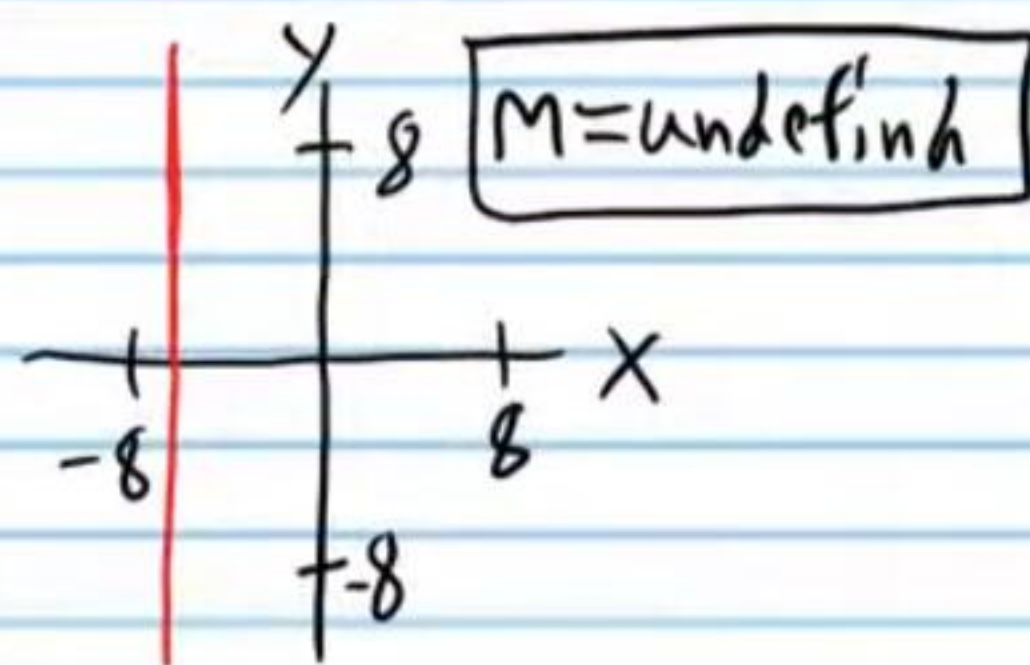
$$\begin{array}{r} -4 \cdot \begin{cases} 3A + 5C = 155 \\ 4A + 2C = 132 \end{cases} \\ 3 \cdot \begin{cases} 3A + 5C = 155 \\ 4A + 2C = 132 \end{cases} \end{array}$$

$$\begin{array}{r} \begin{cases} -12A - 20C = -620 \\ 12A + 6C = 396 \end{cases} \\ \hline 0 - 14C = -224 \end{array}$$

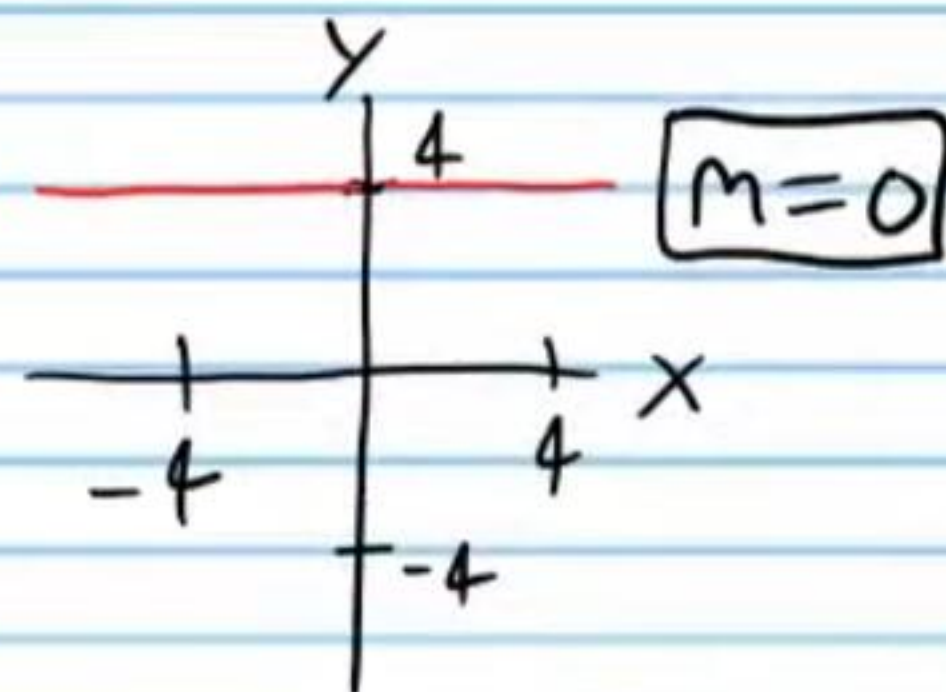
$C = \$16$ per child ticket

⑯ Graph the following equations and identify the slopes. Label axes and intervals.

a) $x = -7$



b) $y = 4$



Solve

$$\textcircled{20} \quad \frac{y}{y^2+6y+8} + \frac{1}{y+4} + \frac{1}{y^2+2y} = \frac{24}{y^3+6y^2+8y}$$

$$\frac{y}{(y+2)(y+4)} + \frac{1}{y+4} + \frac{1}{y(y+2)} = \frac{24}{y(y^2+6y+8)}$$

$$\frac{y}{(y+2)(y+4)} + \frac{1}{y+4} + \frac{1}{y(y+2)} = \frac{24}{y(y+2)(y+4)}$$

$$\frac{y^2}{y(y+2)(y+4)} + \frac{1 \cdot y(y+2)}{(y+4)(y)(y+2)} + \frac{1 \cdot (y+4)}{y(y+2)(y+4)} = \frac{24}{y(y+2)(y+4)}$$

$$\frac{y^2 + y(y+2) + y+4}{y(y+2)(y+4)} = \frac{24}{y(y+2)(y+4)}$$

$$y^2 + y^2 + 2y + y + 4 = 24$$

$$2y^2 + 3y - 20 = 0$$

$$(2y-5)(y+4) = 0$$

$$2y-5=0 \text{ or } y+4=0$$

$$\boxed{y = \frac{5}{2}} \text{ or } \boxed{y = -4}$$

$$\textcircled{21} \quad 2 + \frac{1}{b+5} + \frac{2}{b-2} = \frac{b^2+6b-2}{b^2+3b-10}$$

$$\frac{2}{1} + \frac{1}{b+5} + \frac{2}{b-2} = \frac{b^2+6b-2}{(b+5)(b-2)}$$

$$\frac{2(b+5)(b-2)}{1(b+5)(b-2)} + \frac{1(b-2)}{(b+5)(b-2)} + \frac{2(b+5)}{(b-2)(b+5)} = \frac{b^2+6b-2}{(b+5)(b-2)}$$

$$\frac{2(b^2+3b-10)}{(b+5)(b-2)} + \frac{b-2}{(b+5)(b-2)} + \frac{2b+10}{(b-2)(b+5)} = \frac{b^2+6b-2}{(b+5)(b-2)}$$

$$\frac{2b^2+6b-20+b-2+2b+10}{(b+5)(b-2)} = \frac{b^2+6b-2}{(b+5)(b-2)}$$

$$2b^2+9b-12 = b^2+6b-2$$

$$b^2+3b-10=0$$

$$(b+5)(b-2)=0$$

$$b+5=0 \text{ or } b-2=0$$

$$b \neq -5 \text{ or } b \neq 2$$

No Solutions

$\textcircled{22}$ Bill has 32 coins in his pocket, only quarters and dimes. If the total value of these coins is \$5.90, how many of each type of coin does he have?

$Q = \#$ of quarters

$D = \#$ of dimes

$$\begin{cases} Q + D = 32 \\ 0.25Q + 0.1D = 5.90 \end{cases}$$

$$\begin{aligned} -0.1 \sum Q + D &= 32 \\ \sum 0.25Q + 0.1D &= 5.90 \end{aligned}$$

$$\begin{aligned} \sum -0.1Q - 0.1D &= -3.2 \\ \sum 0.25Q + 0.1D &= 5.90 \end{aligned}$$

$$0.15Q + 0 = 2.70$$

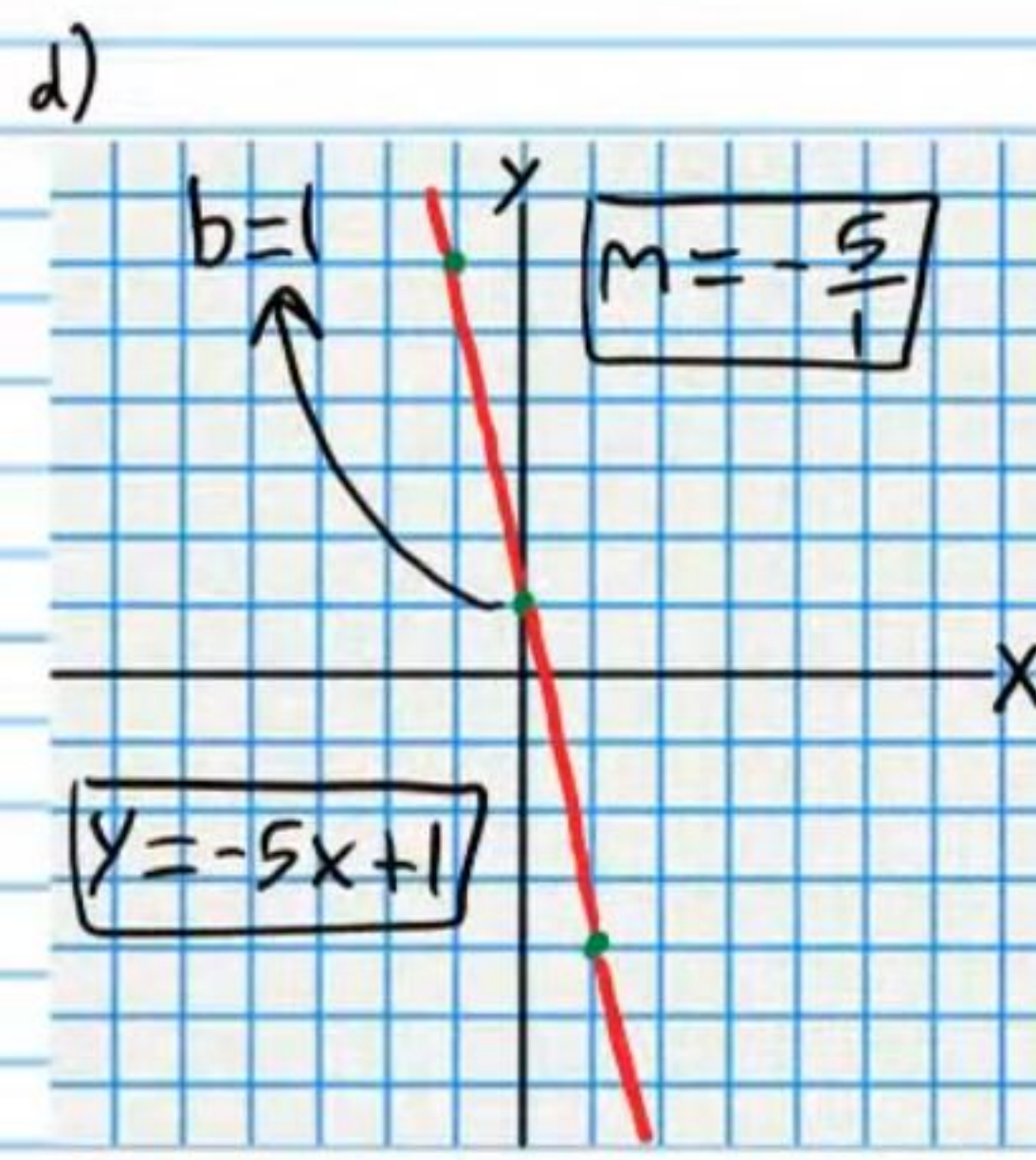
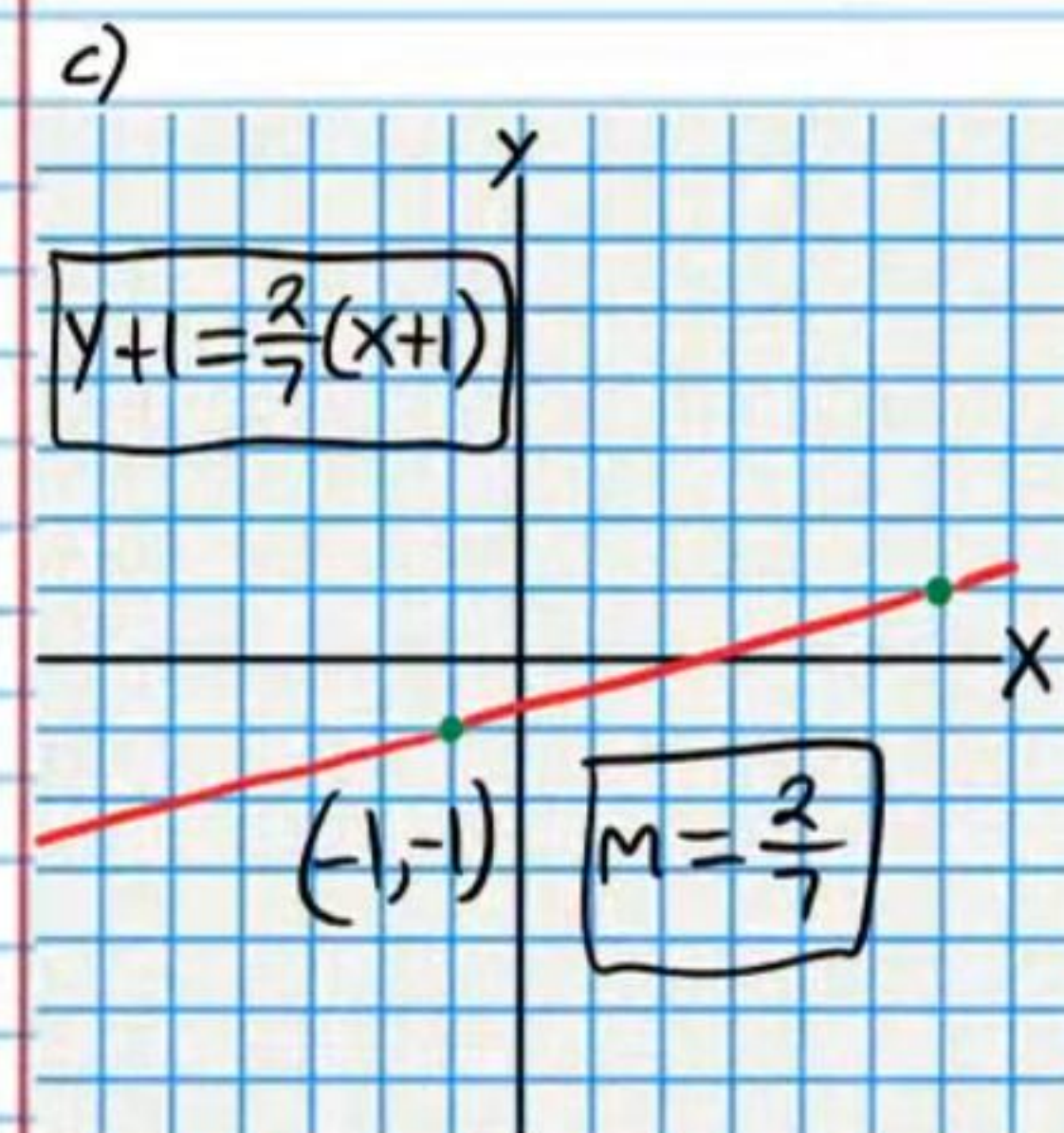
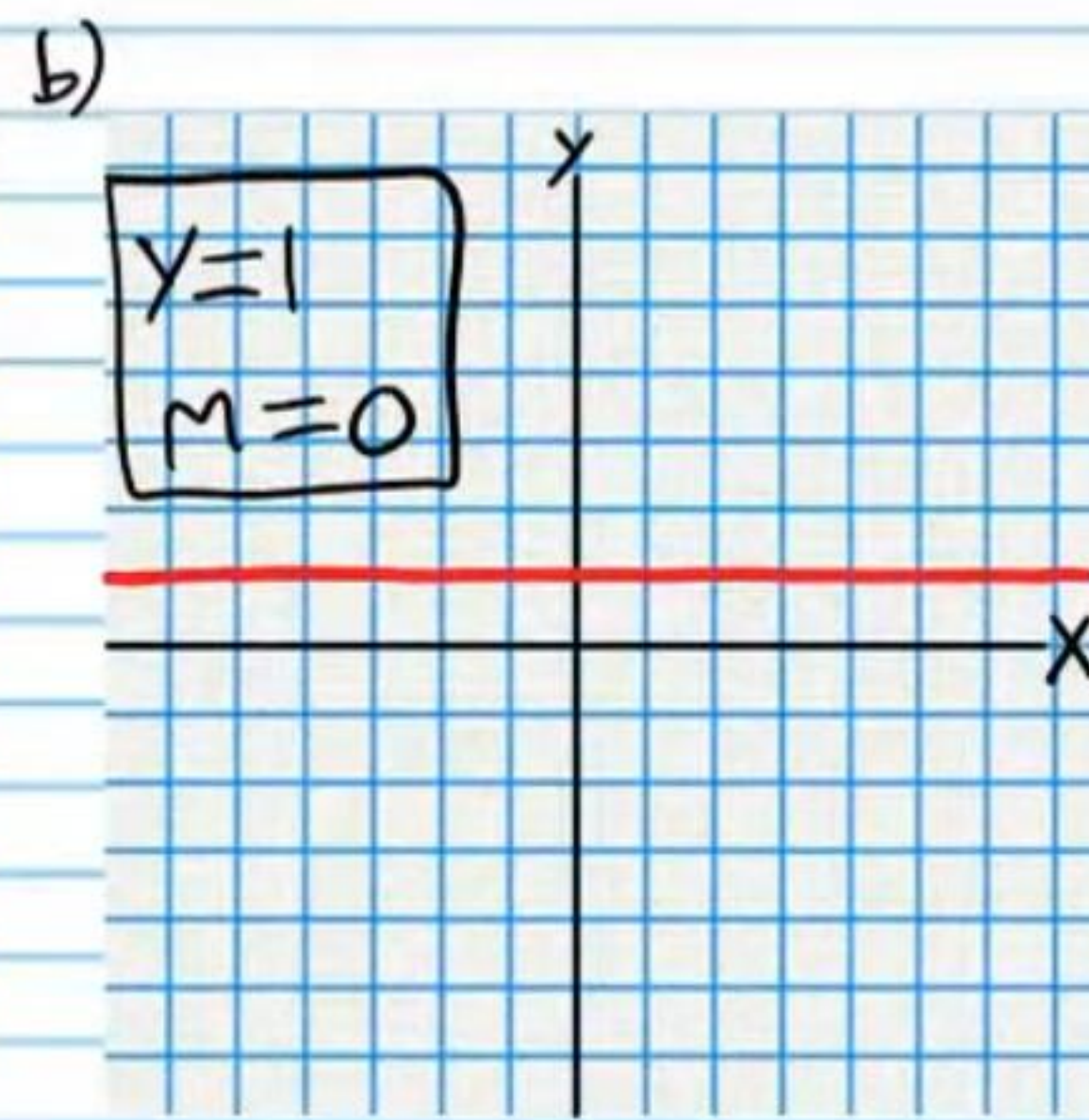
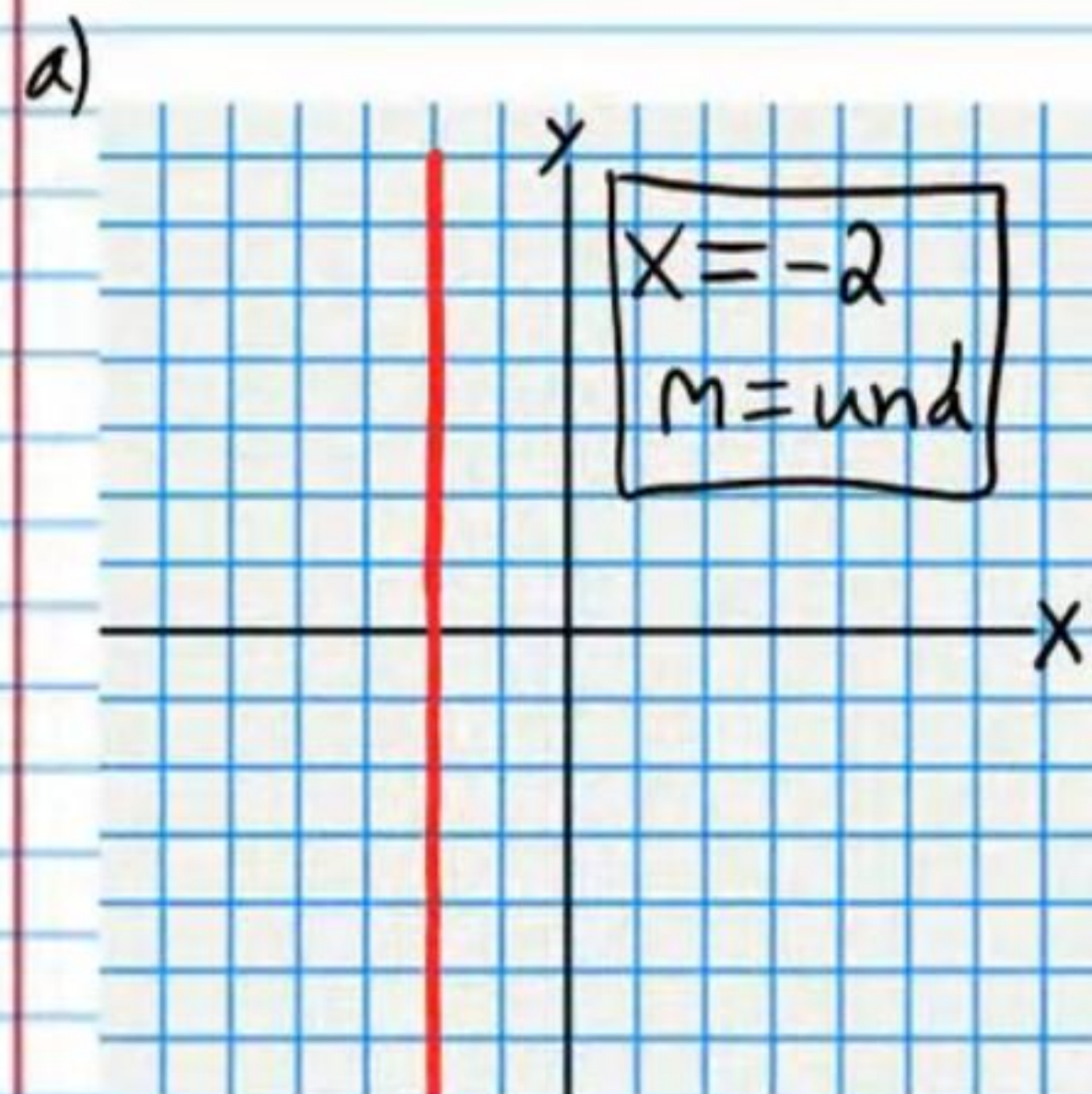
$$0.15Q = 2.7$$

$$Q = 18 \text{ quarters}$$

$$18 + D = 32$$

$$D = 14 \text{ dimes}$$

23) For each graph below, write an equation of the line shown and identify the slope.



24) A bird drops a stick. If we neglect air resistance, the height (in feet) of the stick t seconds after the drop can be modeled with the following equation.

$$h = -16t^2 + 60$$

a) Find the height of the stick one second after the bird drops it.

$$h = -16(1)^2 + 60$$

$$h = -16(1) + 60$$

$$h = -16 + 60$$

$$h = \boxed{44 \text{ ft}}$$

b) How many seconds after the bird drops the stick will it be 30 feet above the ground?

$$30 = -16t^2 + 60$$

$$-30 = -16t^2$$

$$\frac{30}{16} = t^2$$

$$\frac{15}{8} = t^2$$

$$\pm \sqrt{\frac{15}{8}} = \sqrt{t^2}$$

$$\pm \frac{\sqrt{15}}{2\sqrt{2}} = t$$

$$\boxed{\frac{\sqrt{30}}{4} \text{ s} = t}$$

c) How many seconds after the bird drops the stick will it hit the ground?

$$0 = -16t^2 + 60$$

$$-60 = -16t^2$$

$$\frac{-60}{-16} = t^2$$

$$\frac{15}{4} = t^2$$

$$\pm \sqrt{\frac{15}{4}} = \sqrt{t^2}$$

$$\boxed{\frac{\sqrt{15}}{2} s = t}$$

d) How many seconds after the drop will the stick be 10 feet above the ground?

$$10 = -16t^2 + 60$$

$$-50 = -16t^2$$

$$\frac{50}{16} = t^2$$

$$\frac{25}{8} = t^2$$

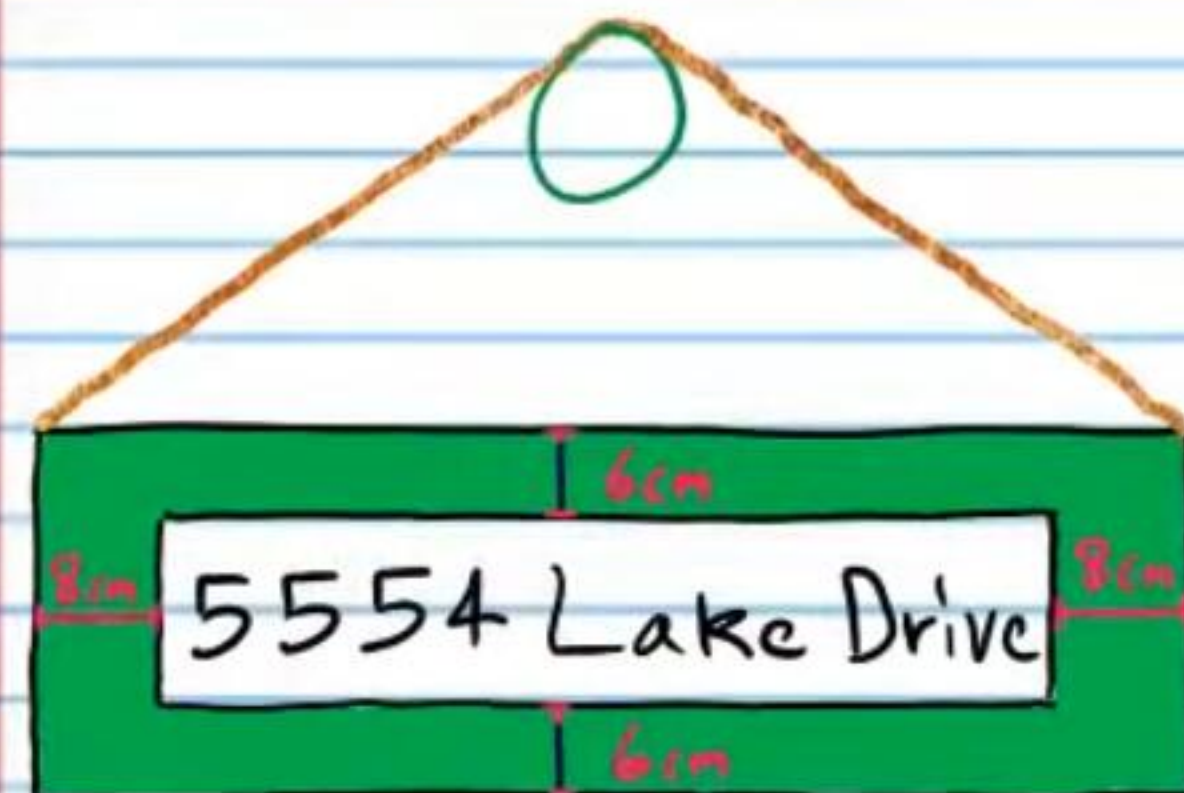
$$\pm \sqrt{\frac{25}{8}} = \sqrt{t^2}$$

$$\pm \frac{5}{\sqrt{8}} = t$$

$$\pm \frac{5}{2\sqrt{2}} = t$$

$$\boxed{\frac{5\sqrt{2}}{4} s = t}$$

25) Mr. Benson's address is written on a flat rectangular sign hanging from his porch. The numbers are written on a smaller rectangle at the center of the sign. The area of the entire face of the sign is 96 square centimeters less than three times the area of the inner rectangle (containing the writing). If the length of the inner rectangle is 28 centimeters shorter than its width, what are the dimensions of the sign?



$$A_T = 3A_I - 96$$

$$(x+8+8)(x-28+6+6) = 3[x(x-28)] - 96$$

$$(x+16)(x-16) = 3x(x-28) - 96$$

$$x^2 - 16x + 16x - 256 = 3x^2 - 84x - 96$$

$$0 = \frac{2x^2 - 84x + 160}{2}$$

$$0 = x^2 - 42x + 80$$

$$0 = (x-40)(x-2)$$

$$x-40=0 \text{ or } x-2=0$$

$$x=40$$

or

$$x \neq 2$$

$$L = 40 - 28 + 6 + 6$$

$$= 24 \text{ cm}$$

$$W = 40 + 8 + 8$$

$$= 56 \text{ cm}$$